

7.14 (a) The Maxwell equation is $\nabla \times (\nabla \times \vec{E}) + \mu_0 \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0$, or

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} + \mu_0 \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0.$$

From $\nabla \cdot \vec{D} = \nabla \cdot (\epsilon \vec{E}) = \nabla \epsilon \cdot \vec{E} + \epsilon \nabla \cdot \vec{E} = 0$, we have $\nabla \cdot \vec{E} = -\frac{1}{\epsilon} \nabla \epsilon \cdot \vec{E}$. Then, the Maxwell equation

becomes

$$-\nabla \left(\frac{1}{\epsilon} \nabla \epsilon \cdot \vec{E} \right) - \nabla^2 \vec{E} + \mu_0 \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0.$$

For horizontal polarization, $\vec{E} = \vec{e}_y E_y(z) e^{ikx - i\omega t}$, $\nabla \epsilon \cdot \vec{E} = 0$, and we will obtain familiar result,

$$\frac{d^2 E_y}{dz^2} + q(z)^2 E_y = 0, \quad \text{where } q(z)^2 = \omega^2 \mu_0 \epsilon(z) - k^2.$$

For vertical polarization, $\vec{E} = \vec{e}_z E_z(z) e^{ikx - i\omega t}$, $\nabla \epsilon \cdot \vec{E} = E_z \frac{d\epsilon}{dz}$. The z-component of the Maxwell equation becomes

$$-\frac{d}{dz} \left(\frac{E_z}{\epsilon} \frac{d\epsilon}{dz} \right) - \frac{d^2 E_z}{dz^2} + k^2 E_z - \omega^2 \mu_0 \epsilon E_z = 0.$$

Let $F = \sqrt{\epsilon/\epsilon_0} E_z$, then

$$\begin{aligned} \frac{d^2 E_z}{dz^2} &= \sqrt{\epsilon_0} \frac{d^2}{dz^2} \left(\frac{F}{\sqrt{\epsilon}} \right) = \sqrt{\epsilon_0} \frac{d}{dz} \left(\frac{1}{\epsilon^{3/2}} \frac{dF}{dz} - \frac{F}{2\epsilon^{5/2}} \frac{d\epsilon}{dz} \right) \\ &= \sqrt{\epsilon_0} \left(\frac{1}{\epsilon^{5/2}} \frac{d^2 F}{dz^2} - \frac{1}{\epsilon^{3/2}} \frac{dF}{dz} \frac{d\epsilon}{dz} + \frac{3F}{4\epsilon^{5/2}} \left(\frac{d\epsilon}{dz} \right)^2 - \frac{F}{2\epsilon^{3/2}} \frac{d^2 \epsilon}{dz^2} \right) \end{aligned}$$

$$\frac{d}{dz} \left(\frac{E_z}{\epsilon} \frac{d\epsilon}{dz} \right) = \sqrt{\epsilon_0} \frac{d}{dz} \left(\frac{F}{\epsilon^{3/2}} \frac{d\epsilon}{dz} \right) = \sqrt{\epsilon_0} \left(\frac{1}{\epsilon^{3/2}} \frac{dF}{dz} \frac{d\epsilon}{dz} - \frac{3F}{2\epsilon^{5/2}} \left(\frac{d\epsilon}{dz} \right)^2 + \frac{F}{\epsilon^{3/2}} \frac{d^2 \epsilon}{dz^2} \right),$$

and the Maxwell equation becomes

$$\sqrt{\epsilon_0} \left(\frac{1}{\epsilon^{5/2}} \frac{d^2 F}{dz^2} + \frac{F}{2\epsilon^{3/2}} \frac{d^2 \epsilon}{dz^2} - \frac{3F}{4\epsilon^{5/2}} \left(\frac{d\epsilon}{dz} \right)^2 \right) + (\omega^2 \mu_0 \epsilon - k^2) E_z = 0,$$

or

$$\frac{d^2 F}{dz^2} + q(z)^2 F = 0, \quad \text{with } q(z)^2 = \omega^2 \mu_0 \epsilon + \frac{1}{2\epsilon} \frac{d^2 \epsilon}{dz^2} - \frac{3}{4\epsilon^2} \left(\frac{d\epsilon}{dz} \right)^2 - k^2.$$

(b) From Eq. (7.59), $q(z) = \epsilon_0 (1 - \omega_p(z)^2/\omega^2)$, and the equation for vertical polarization

becomes

$$\frac{d^2 F}{dz^2} + \mu_0 \epsilon_0 (\omega^2 - \omega_p(z)^2) F = 0,$$

dropping slowly varying terms. When $\omega > \omega_p(z)$, there is a travelling solution to the

equation. For $\omega_p(z) > \omega$, the solution is a decaying one. Therefore, there will be reflection at turning point. $\omega_p(z) = \omega$.

(c) We can write the time T as

$$T = 2 \int_0^{h(\omega)} \frac{dz}{\sqrt{\omega^2 - \omega_p(z)^2}} = \frac{zh'(\omega)}{c}.$$

Solution of the above equation gives the effective height.